

PHY 555 Lecture 3

Crystal Structures

- Bravais lattice: simply describes the periodic arrangement of the repeated units of a crystal;
- Crystal: any structure that is made up from a well defined unit structure infinitely repeated following a fixed geometrical rule (periodicity).
- Crystal structure: periodic^a, infinite repetition^b of identical structural units^c.

a. Translational symmetry:

Def. Bravais Lattice= all points defined by

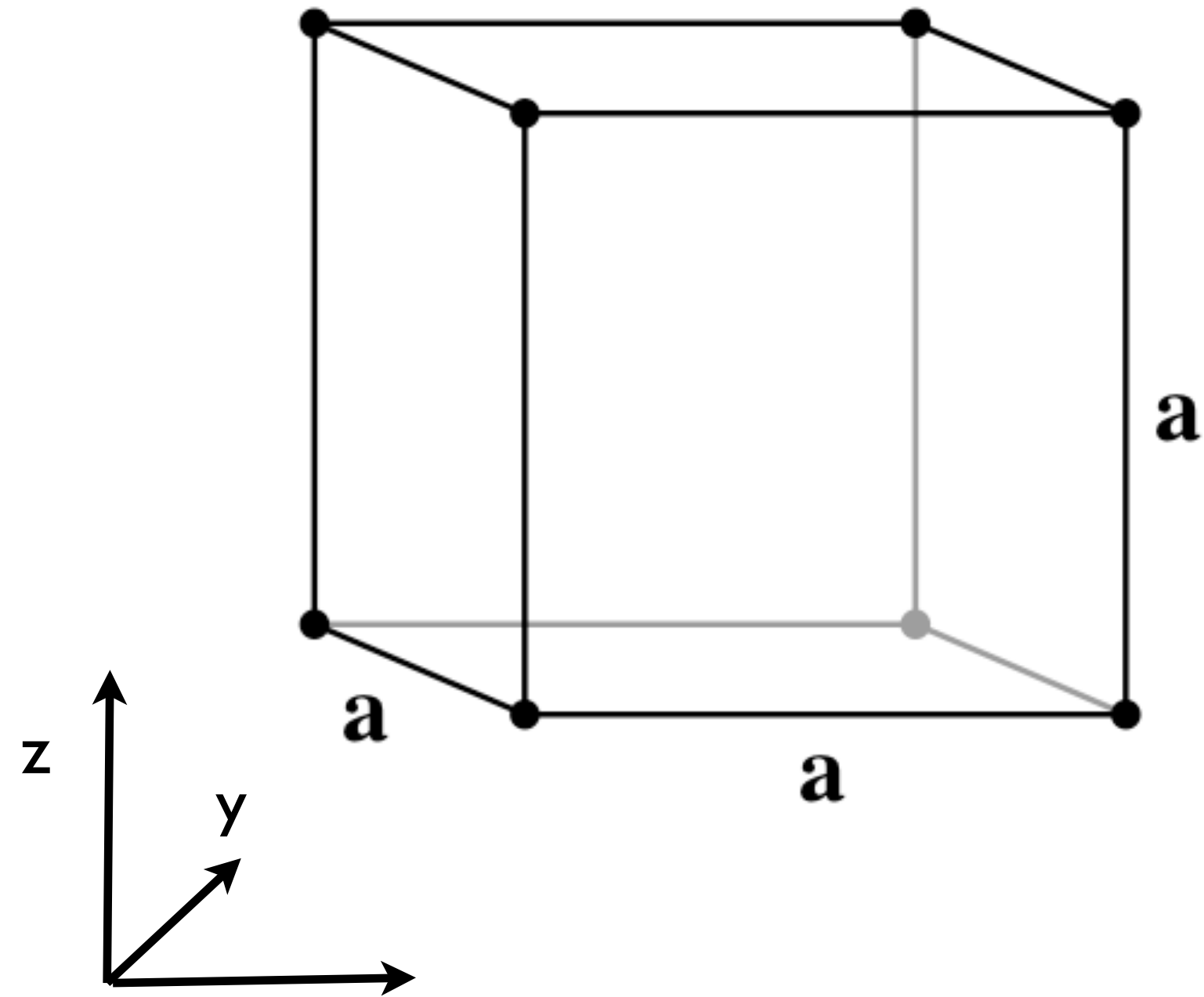
$$\vec{R} \quad (\text{lattice vector}) = n\vec{a}_1 + n\vec{a}_2 + n\vec{a}_3; \quad n_1, n_2, n_3 \text{ integers}$$

$\hat{a}_{1,2,3}$ = primitive lattice vectors (non coplanar) $\Rightarrow$ generate the lattice

b. $N \sim 10^{23} \sim \infty \Rightarrow$ Bulk properties are independent of surfaces

c. Structural units associated to each lattice point $\equiv$ basis = atom (or collection of atoms) or molecules.

Summary: Crystal structure = Bravais lattice + basis

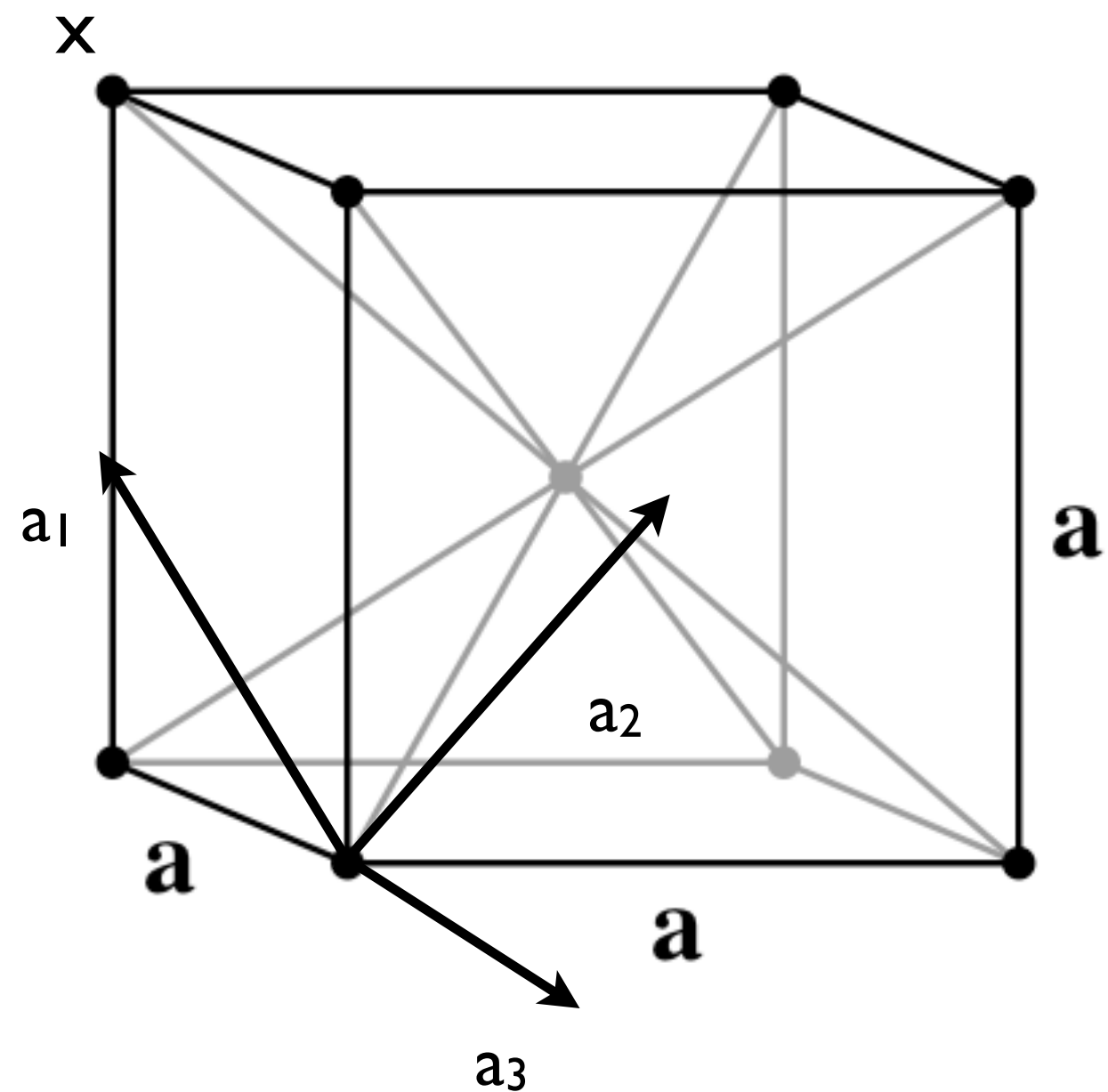


Simple Cubic: SC

$$\vec{a}_1 = a\vec{i}$$

$$\vec{a}_2 = a\vec{j}$$

$$\vec{a}_3 = a\vec{k}$$



Body Centered Cubic: BCC

$$\vec{a}_1 = a\vec{i}$$

$$\vec{a}_2 = a\vec{j}$$

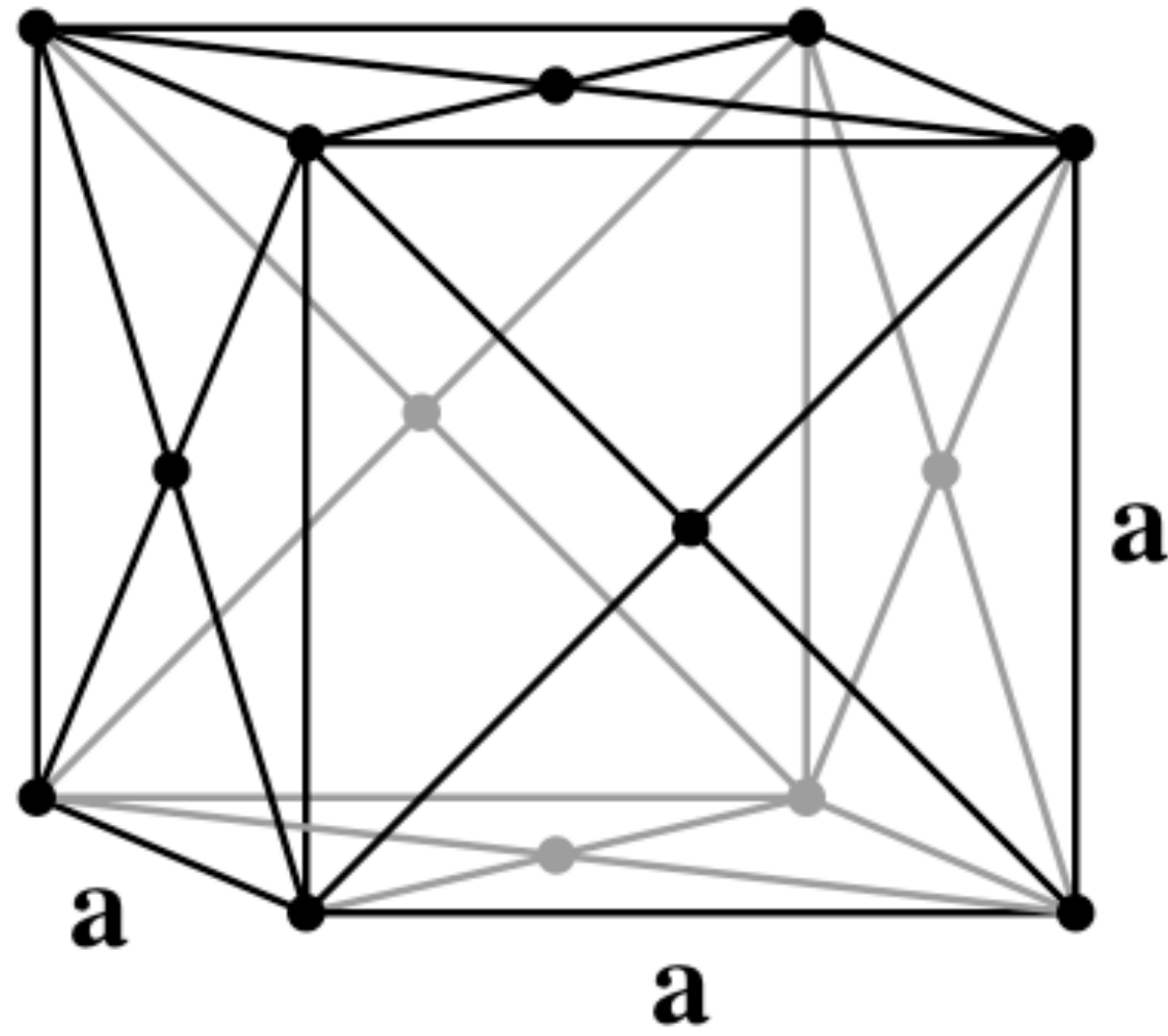
$$\vec{a}_3 = a/2(\vec{i} + \vec{j} + \vec{k})$$

$$\vec{a}_1 = a/2(-\vec{i} + \vec{j} + \vec{k})$$

$$\vec{a}_2 = a/2(\vec{i} - \vec{j} + \vec{k})$$

$$\vec{a}_3 = a/2(\vec{i} + \vec{j} - \vec{k})$$

Face Centered Cubic: FCC



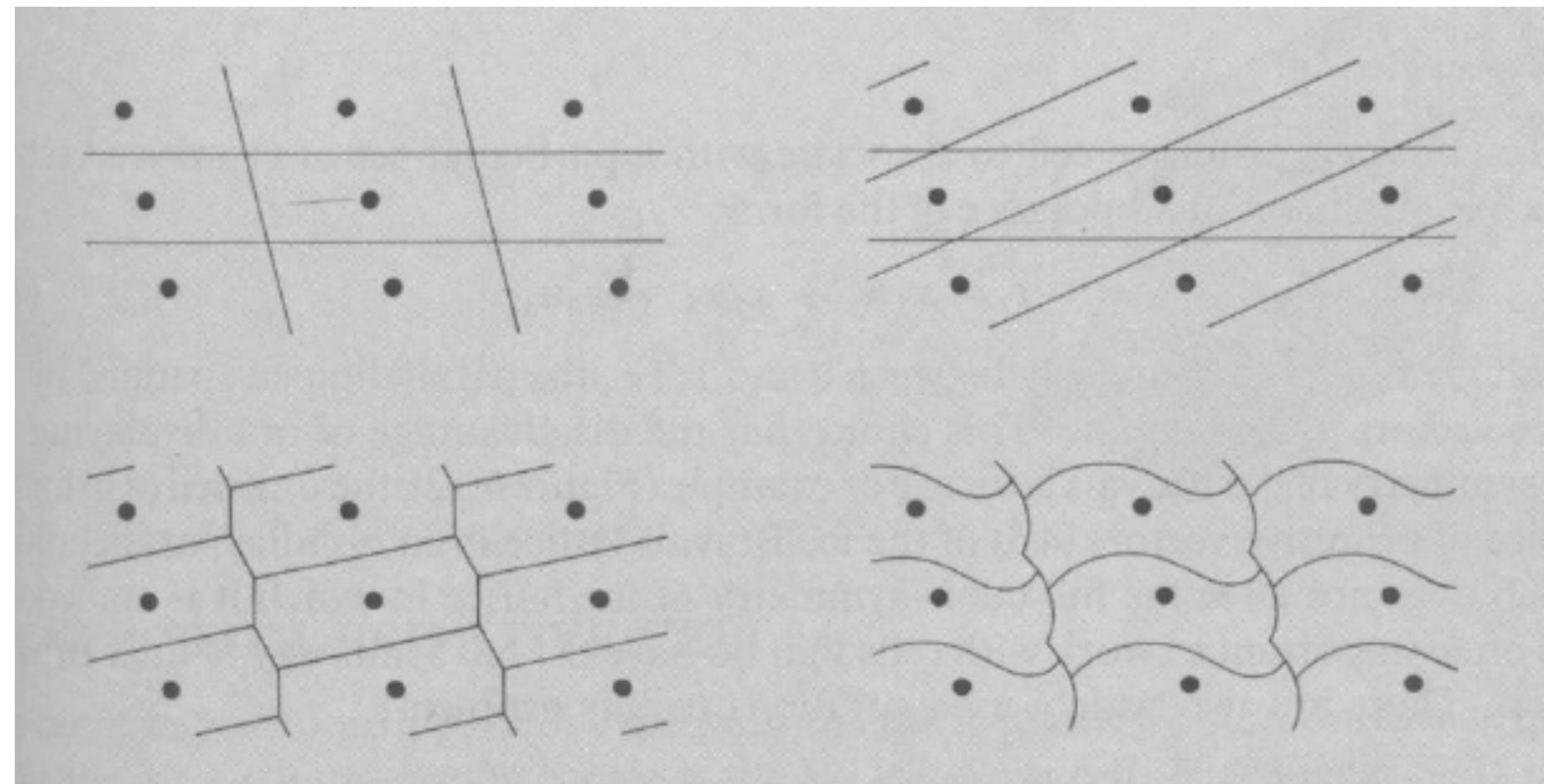
$$\vec{a}_1 = a/2(\vec{j} + \vec{k})$$

$$\vec{a}_2 = a/2(\vec{i} + \vec{k})$$

$$\vec{a}_3 = a/2(\vec{i} + \vec{j})$$

Primitive Unit Cell

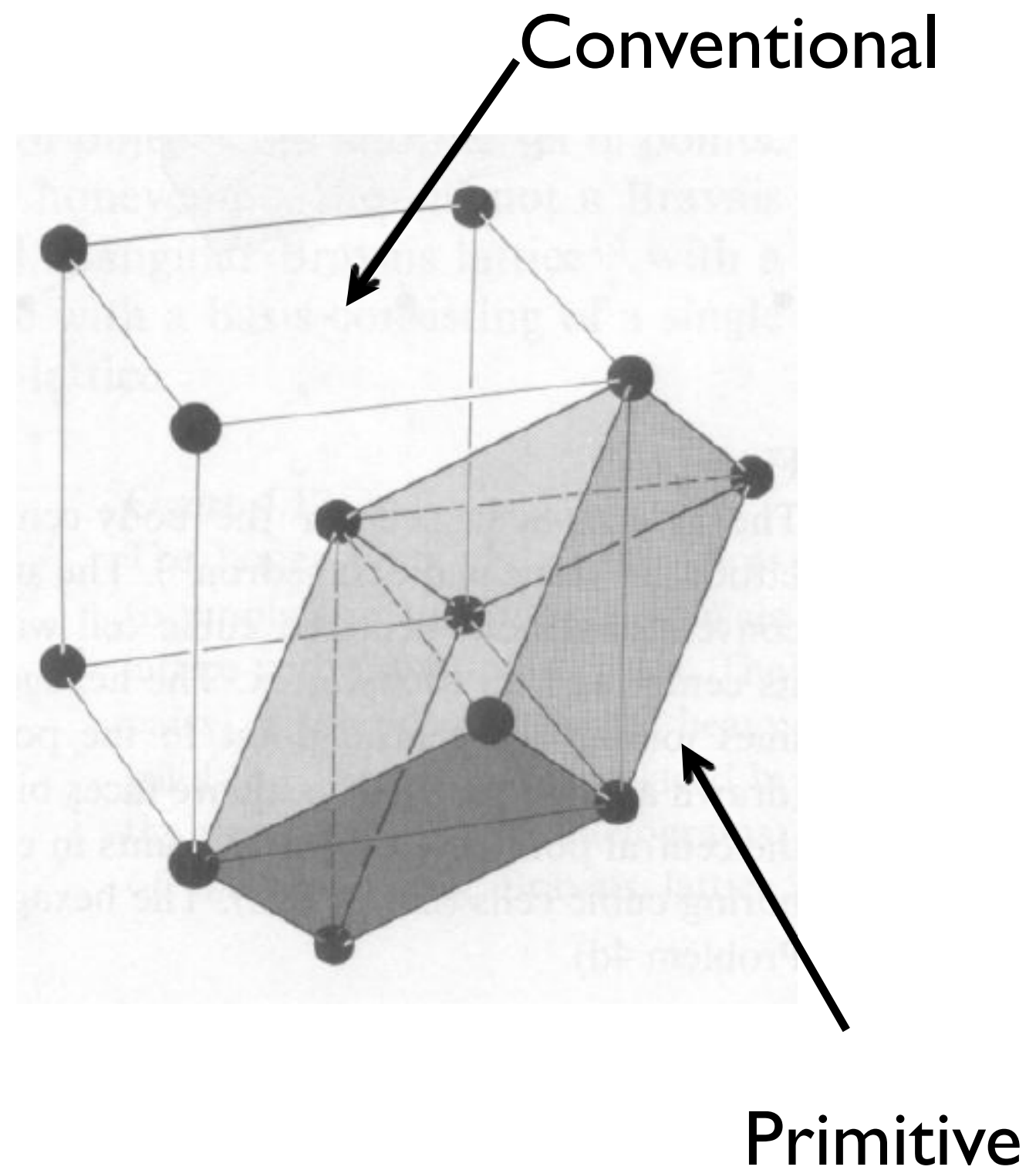
The Space-filling Volume (or hyper-volume) per lattice point can take any arbitrary shape. Whichever volume that when translated through **ALL** vectors in a Bravais lattice fills the full space (no overlapping) represents a primitive unit cell.



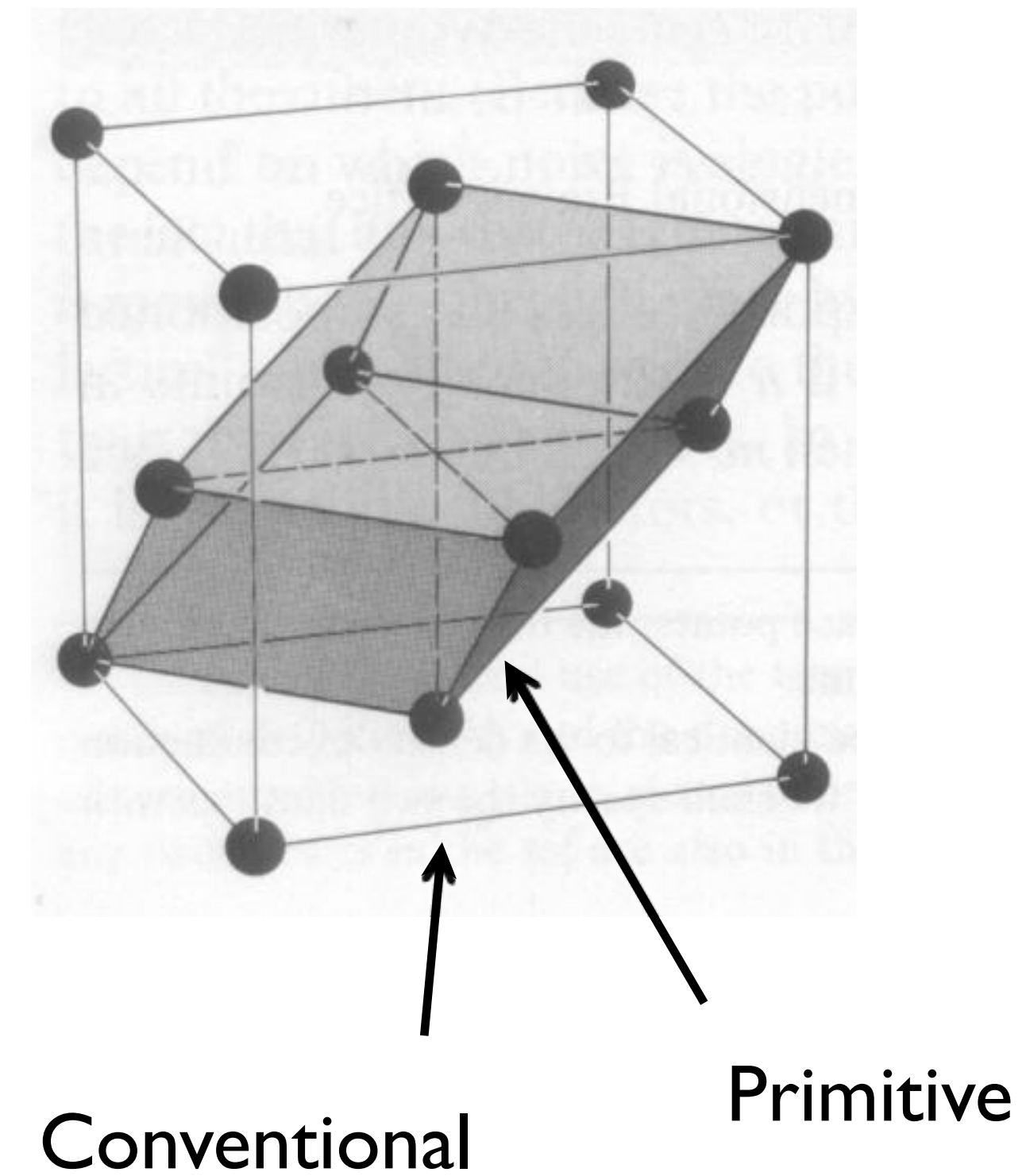
Conventional unit cell

Whichever volume that when translated through a **SUBSET** of vectors in a Bravais lattice fills the full space (no overlapping) represents a primitive unit cell.

bcc

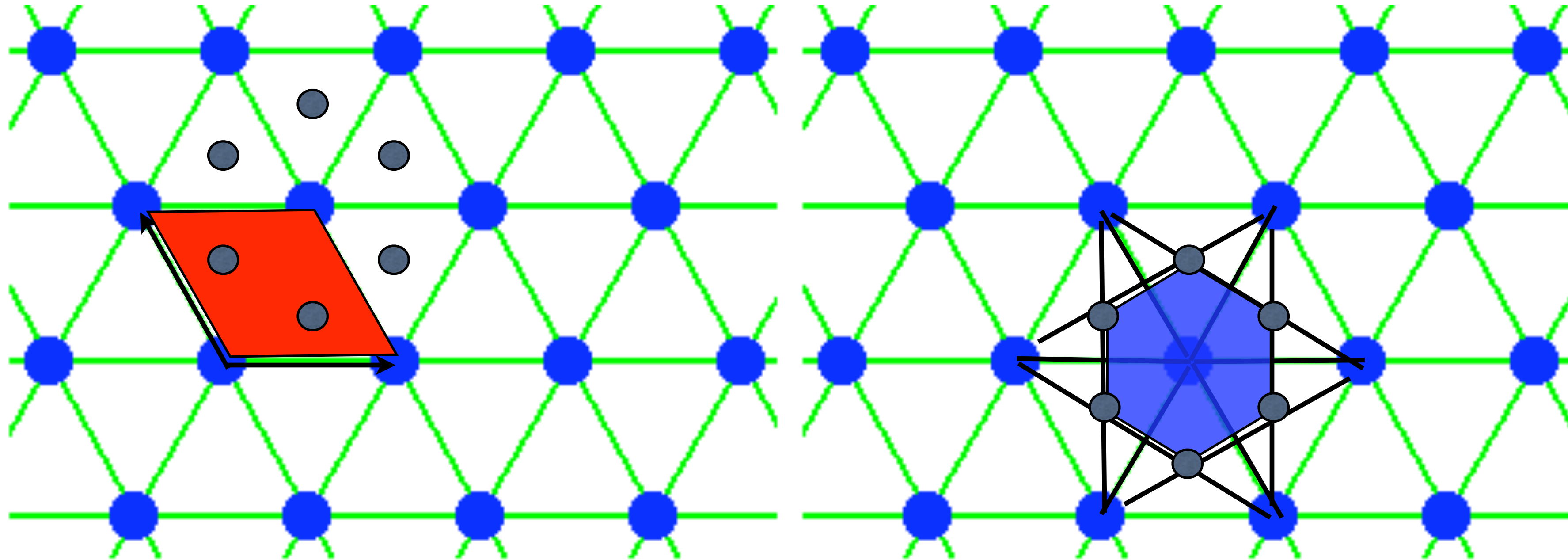


fcc



Unit Cell

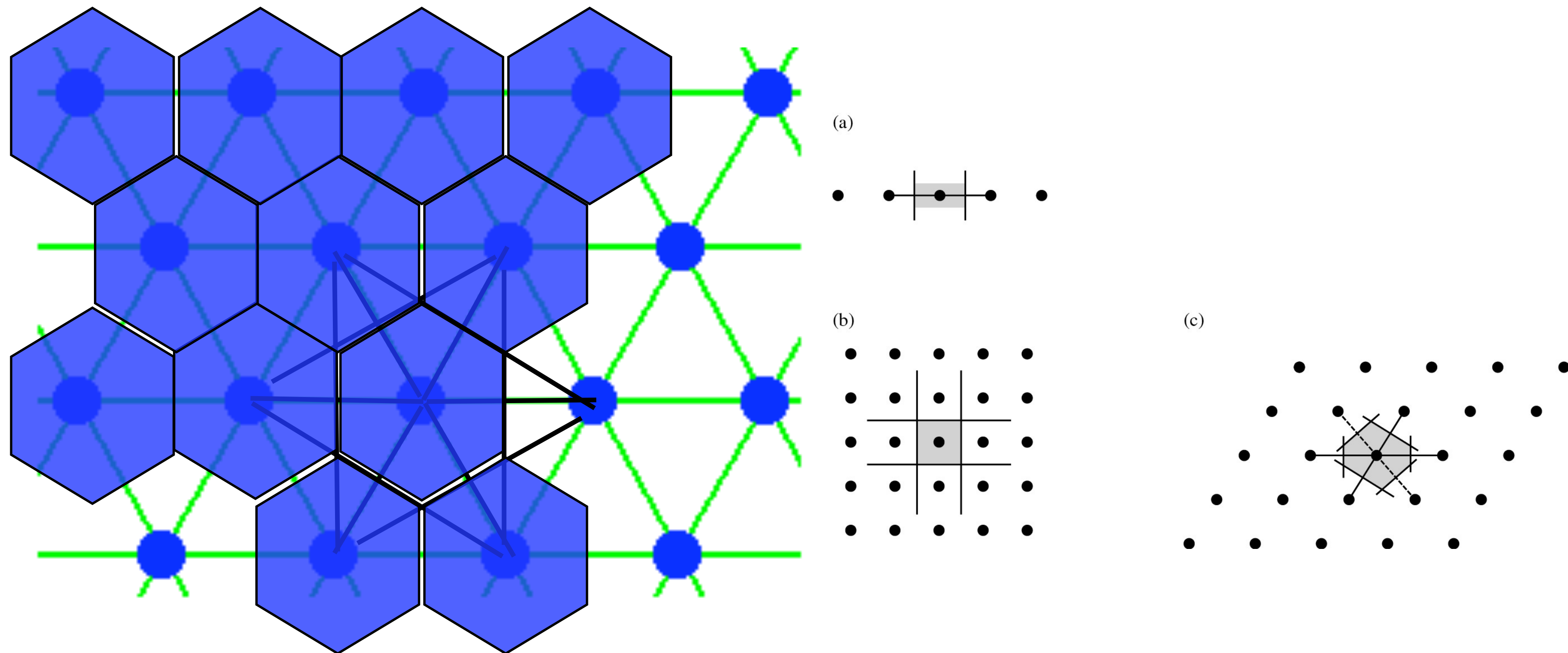
Bravais Lattice (NOT crystal) of graphene: Hexagonal cell



http://home.hetnet.nl/~turing/complete_hex_motif_2a.gif

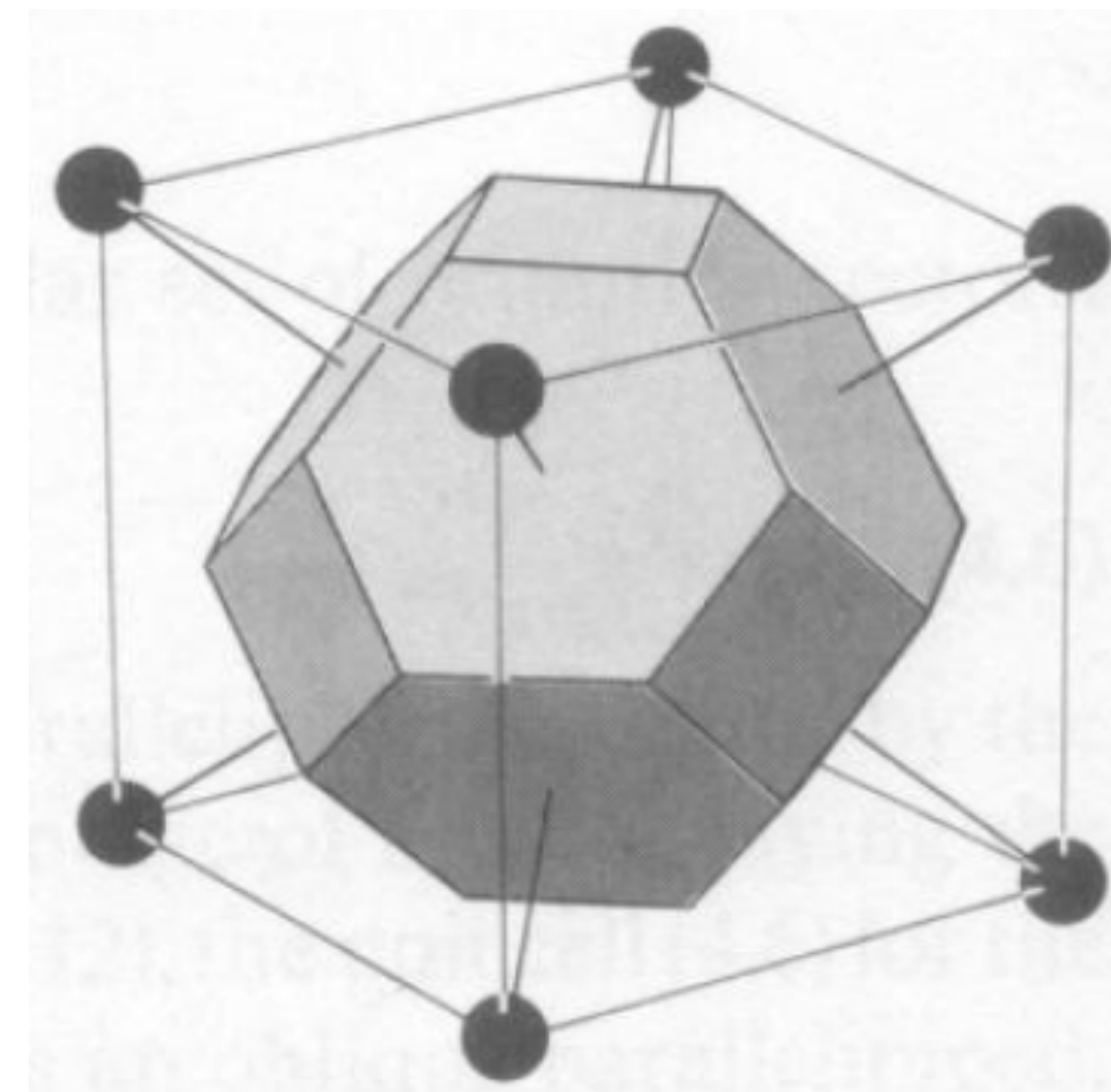
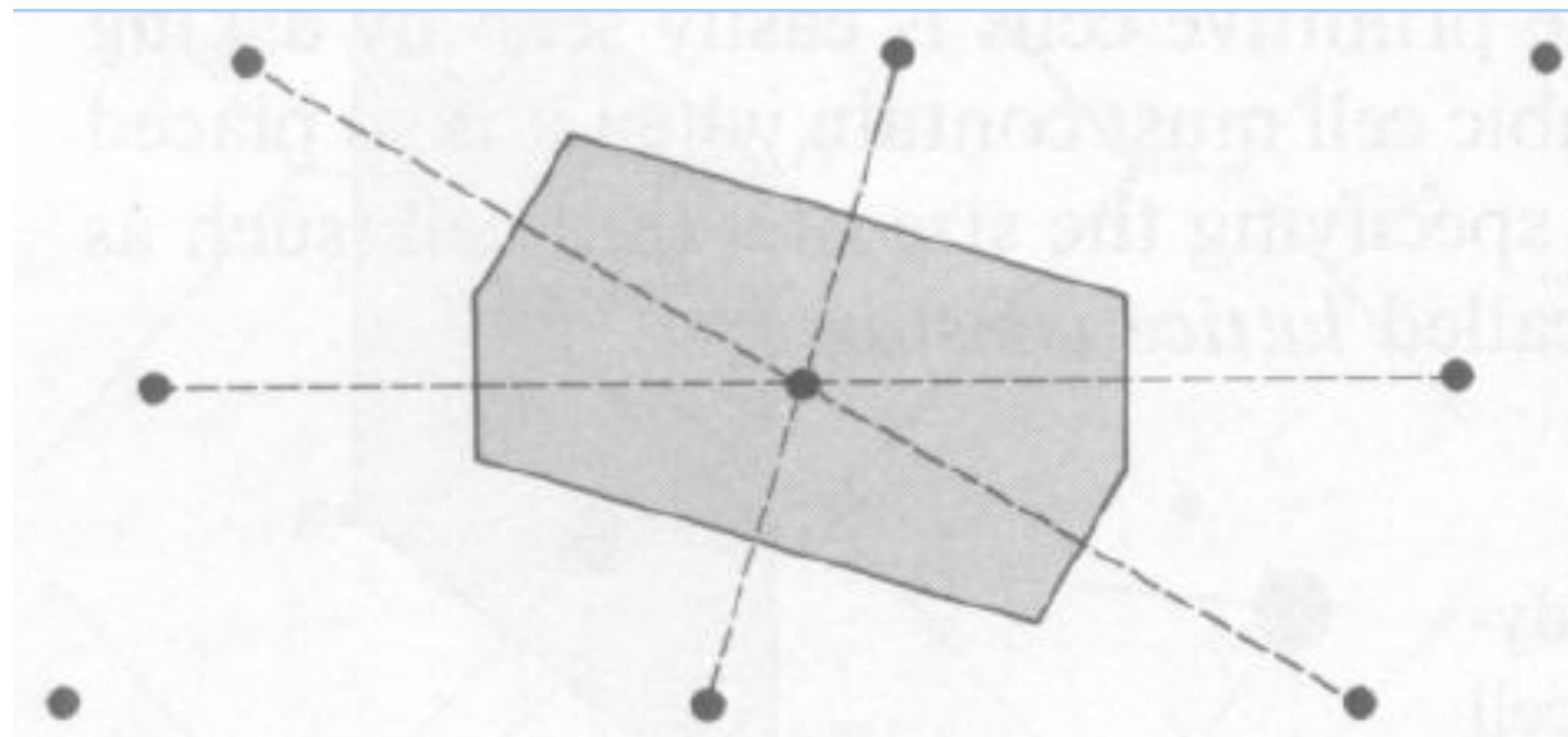
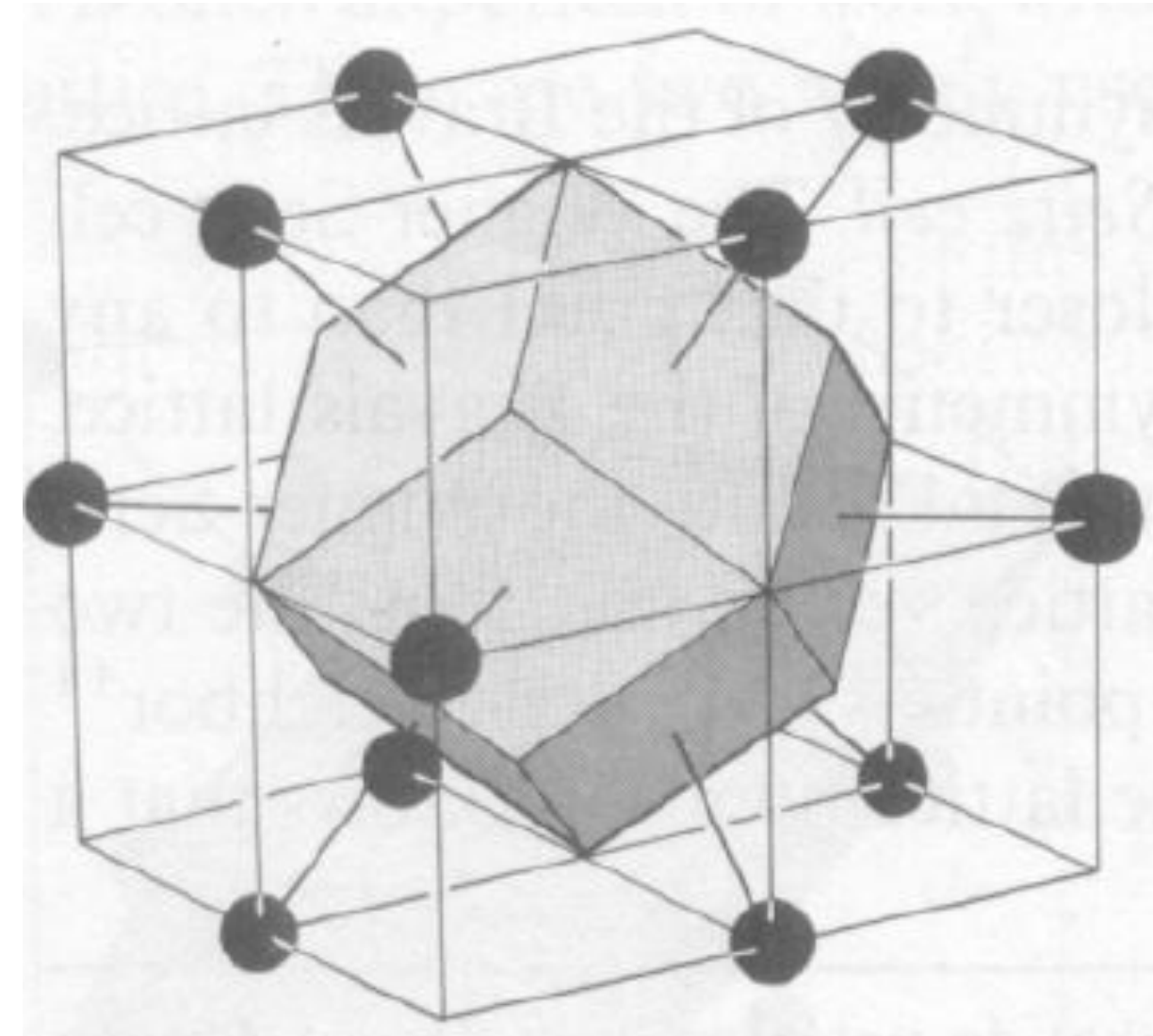
Wigner-Seitz Unit cell

Reflects all (point) symmetries of the Bravais lattice itself.

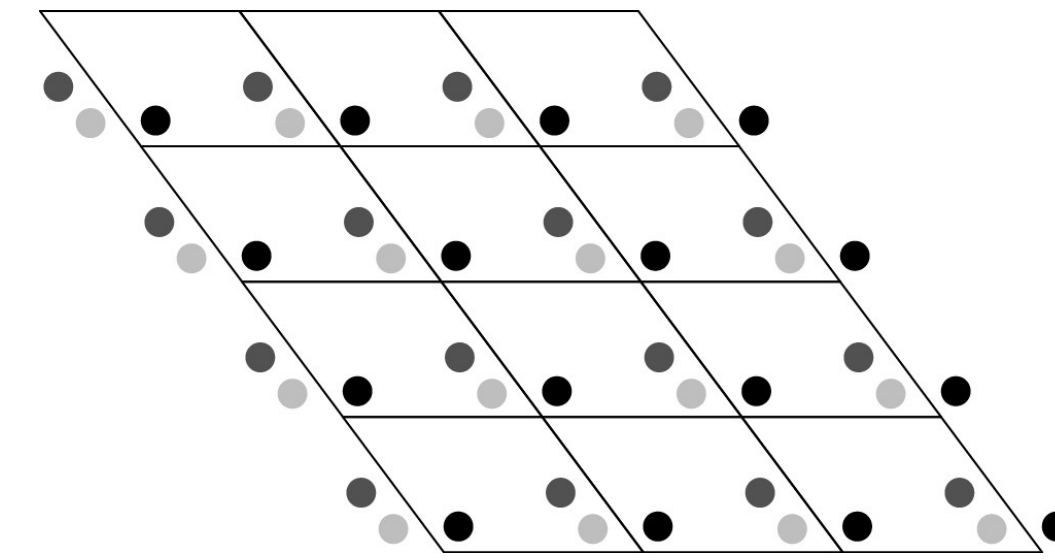


Wigner-Seitz Unit cell

- Region of the space closer to a given lattice point than to any other latt. point.
- Draw Lines from the origin to all nearby lattice points
- Volume enclosed by perpendicular bisector planes
- Shows full Bravais lattice point symmetry

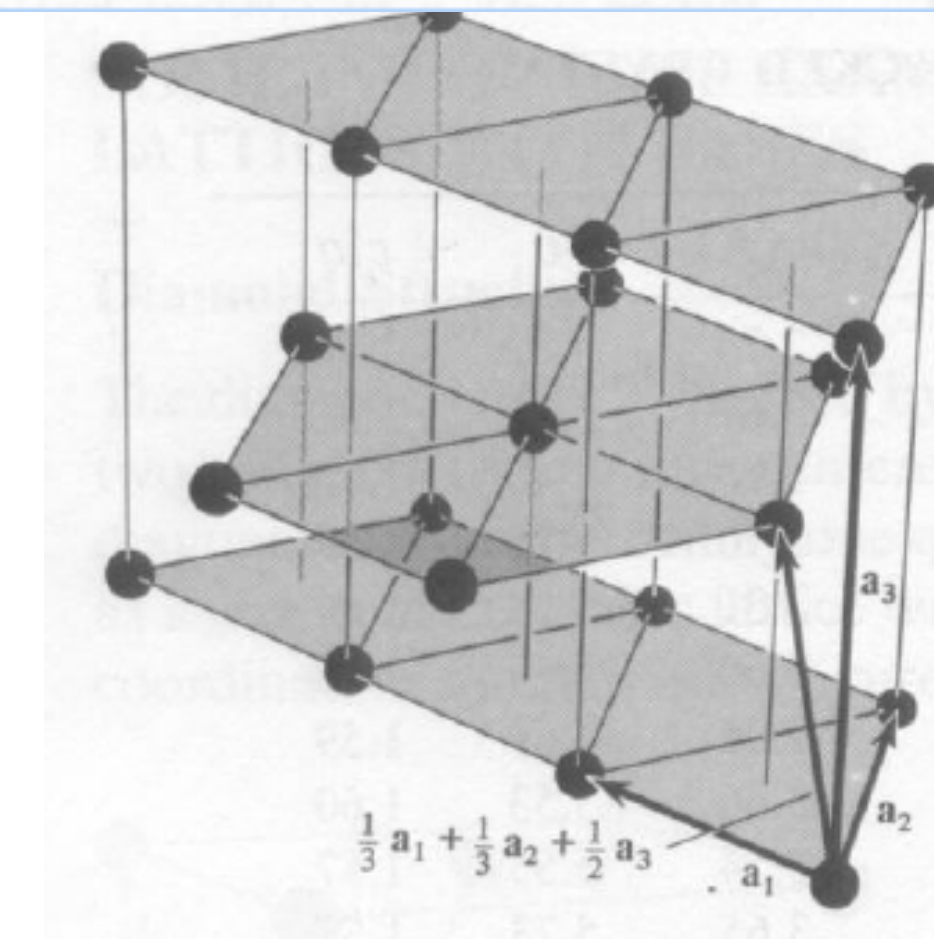
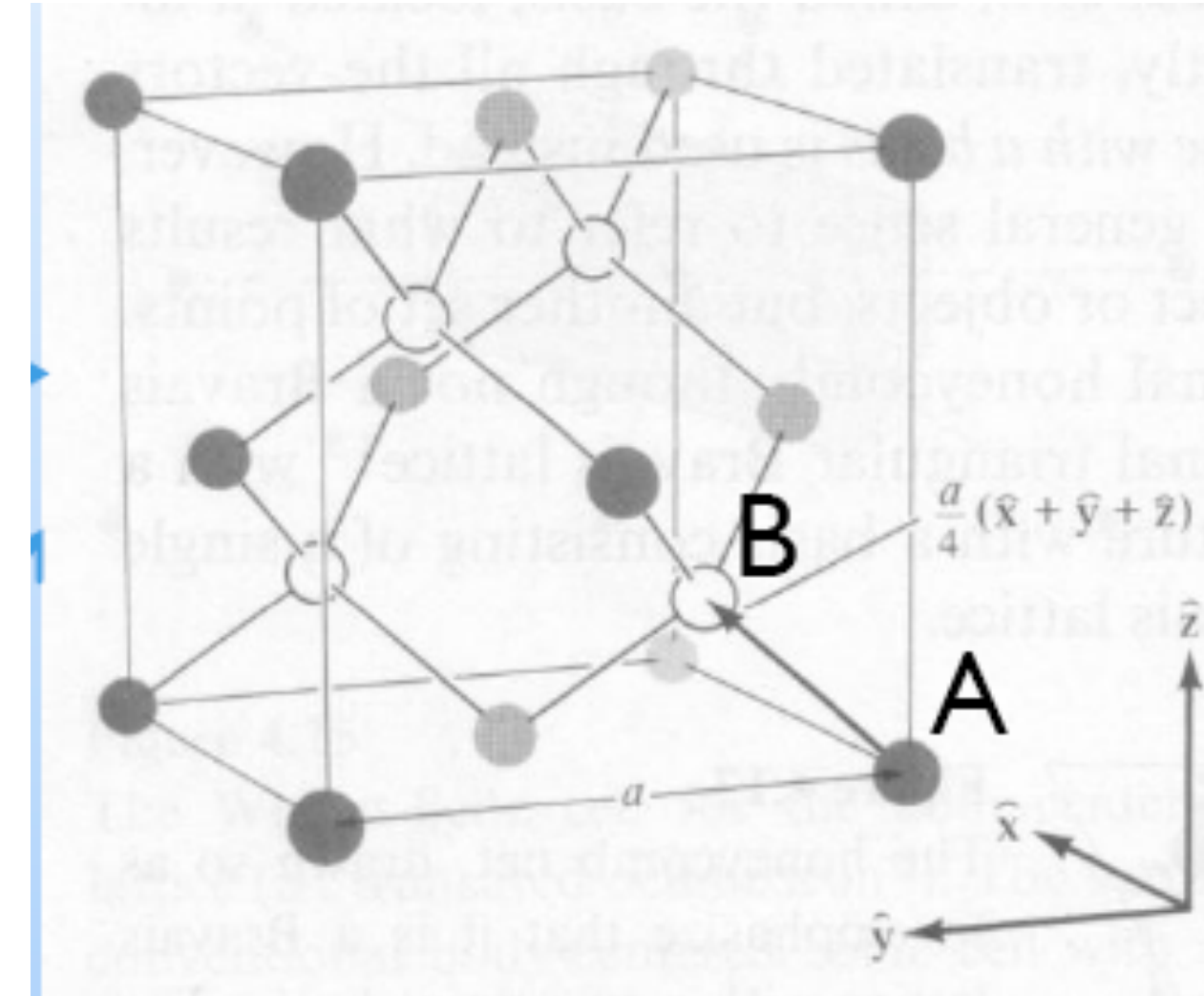


Crystal= Bravais Lattice +Basis



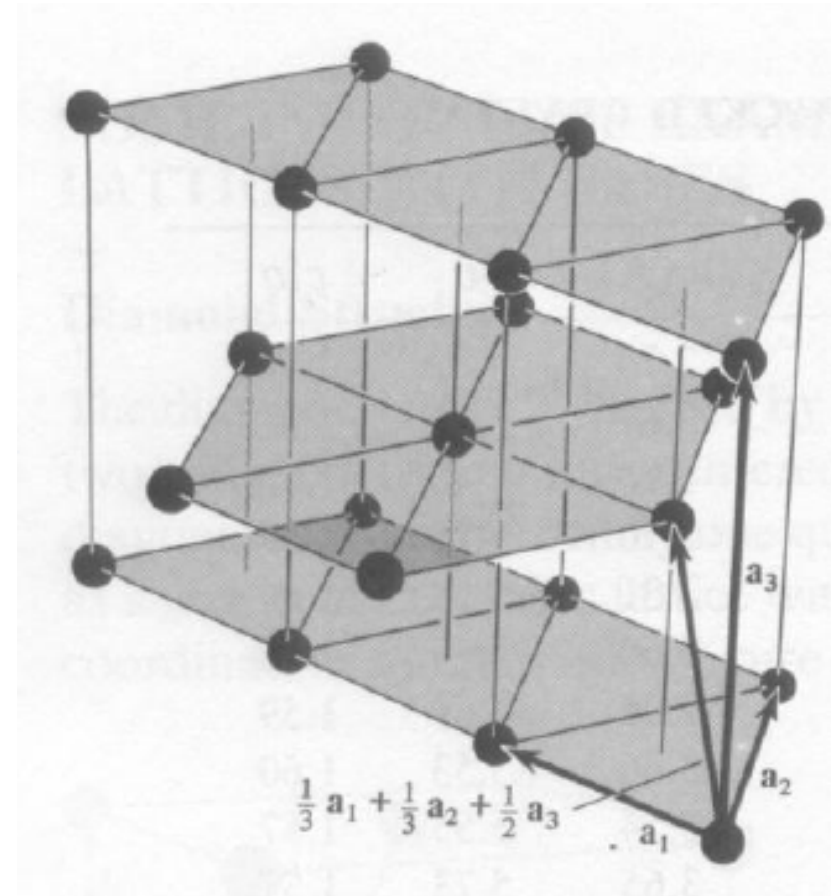
Diamond Structure: Two fcc lattices displaced along body diagonal: $\text{fcc} + \{0, a/4(x+y+z)\}$

- Diamond Structure
(fcc + 2 identical atom basis; $A=B$)
- Zinc-Blend Structure
(fcc + 2 different atom basis; $A \neq B$)
- Hexagonal Close Packed Structure (hcp)
(hexagonal lattice + 2 identical atom basis)

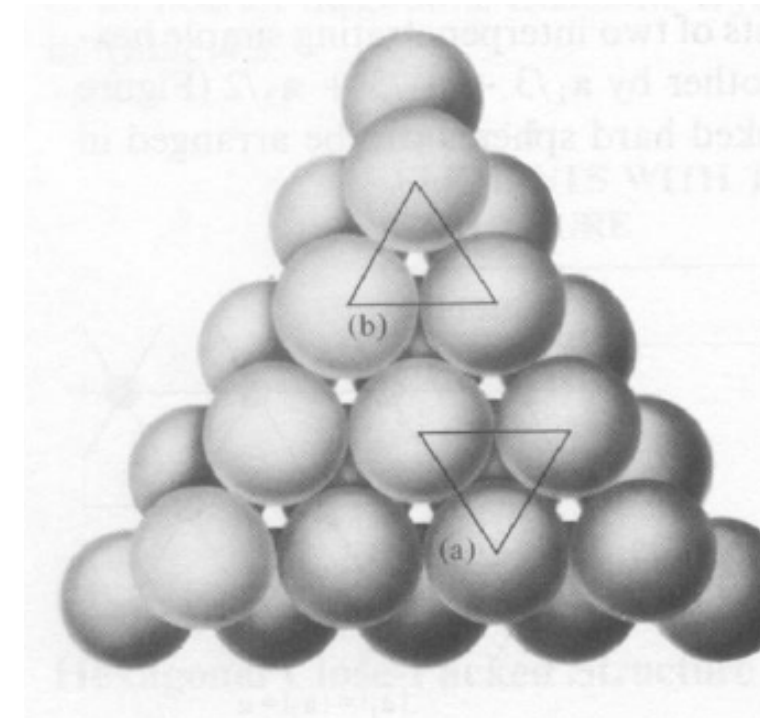


Close-Packed Structures

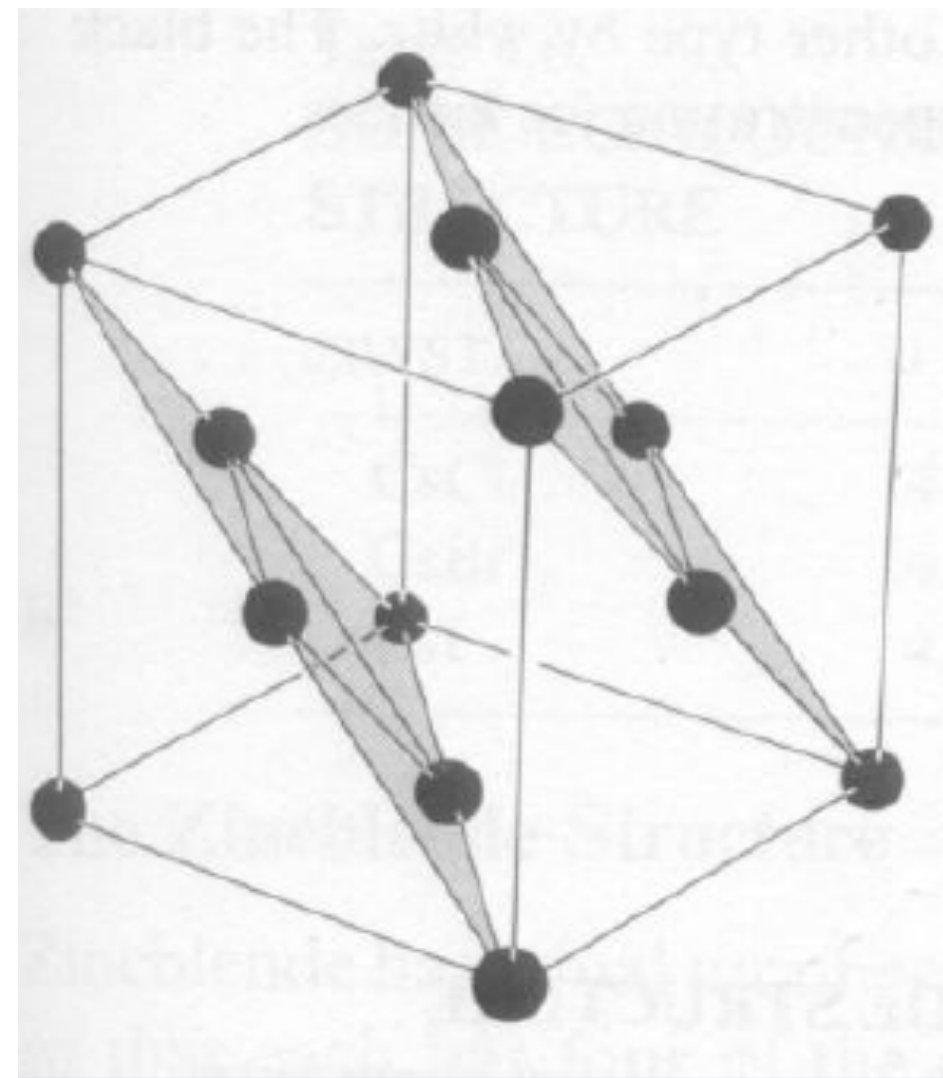
hcp



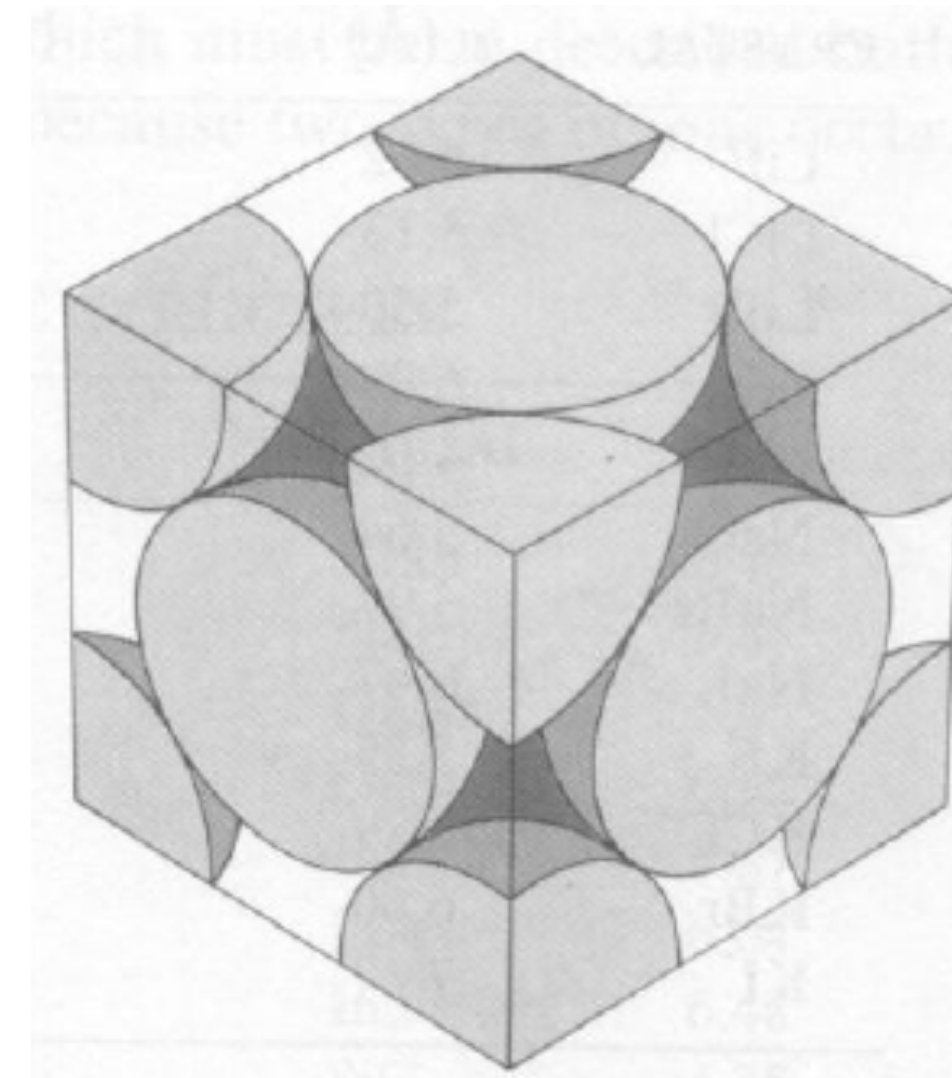
AB-AB packing
of “marbles”



fcc

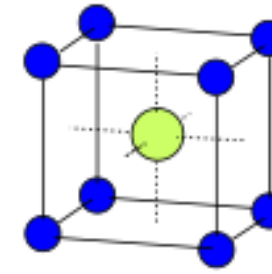


ABC-ABC packing
of “marbles”

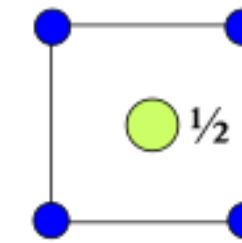


cesium chloride (CsCl)

lattice: cubic P
basis :
Cs 000 ●
Cl $\frac{1}{2}\frac{1}{2}\frac{1}{2}$ ●

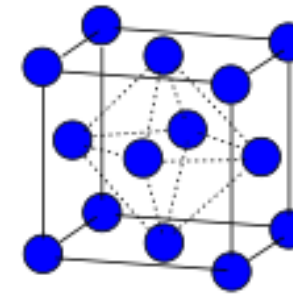


Plan view

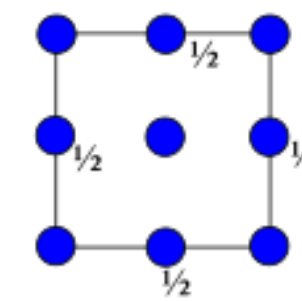


copper (Cu)

lattice: cubic F
basis :
Cu 000 ●

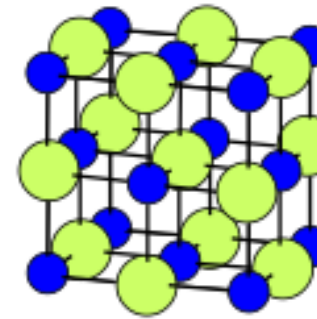


Plan view

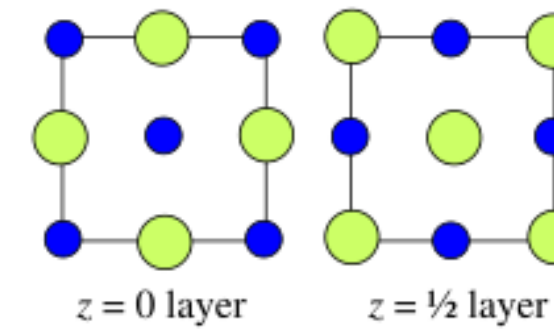


sodium chloride (NaCl)

lattice: cubic F
basis :
Na 000 ●
Cl $\frac{1}{2}\frac{1}{2}\frac{1}{2}$ ●

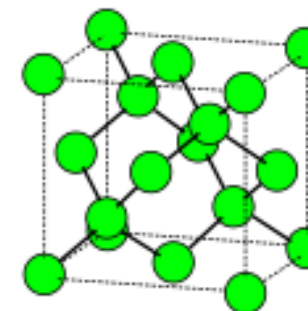


Plan view



diamond (C) — also Si, Ge

lattice: cubic F
basis :
C 000 ●
C $\frac{1}{4}\frac{1}{4}\frac{1}{4}$ ●



Plan view

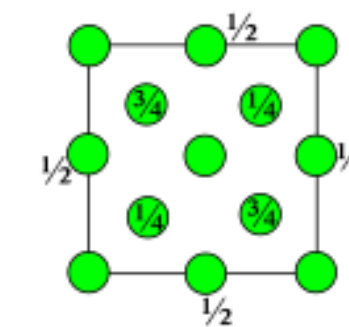


Figure 11.16: Some examples of real crystals with simple structures. Note that in all cases the basis is described with respect to the primitive unit cell of a simple cubic lattice.

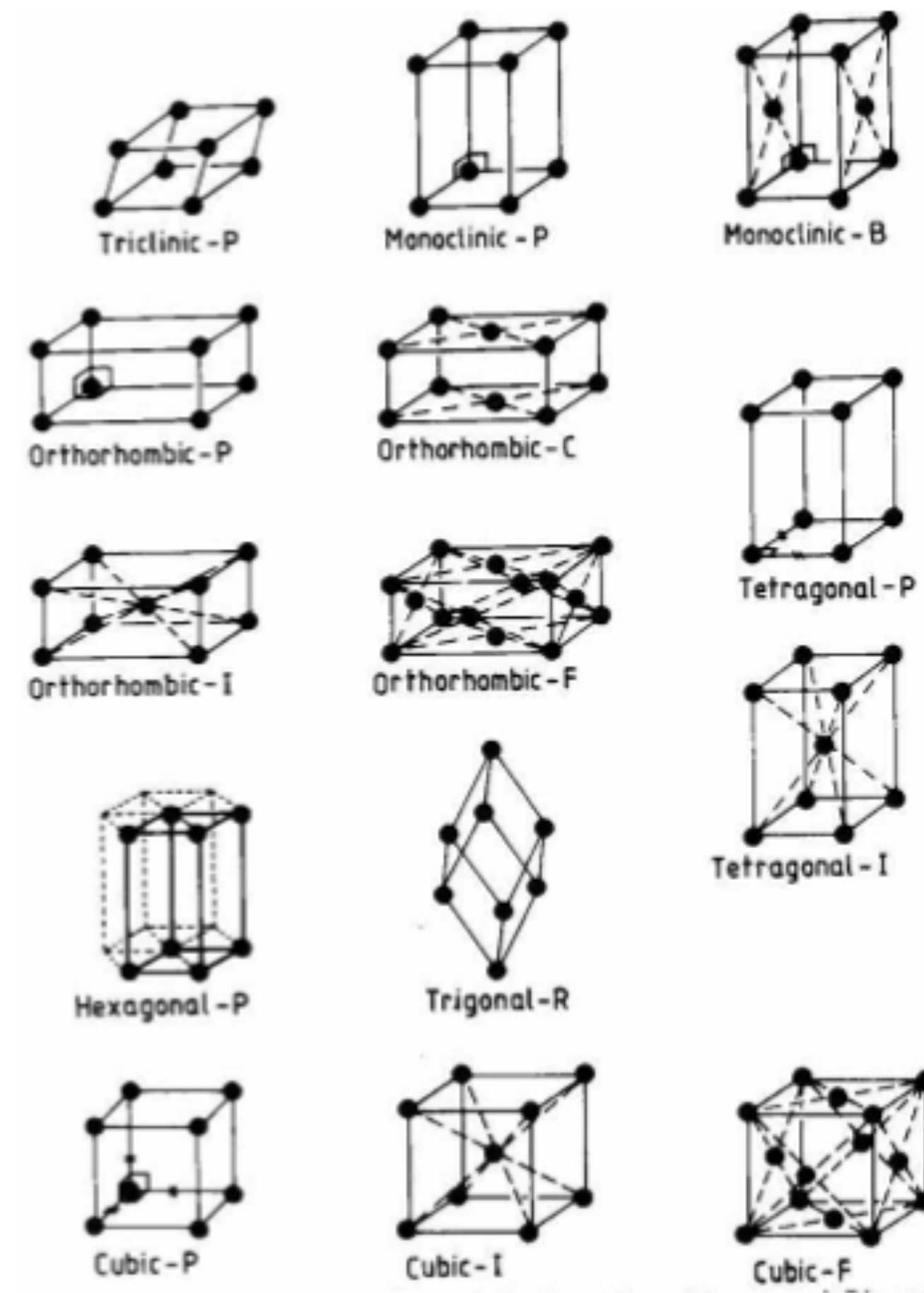
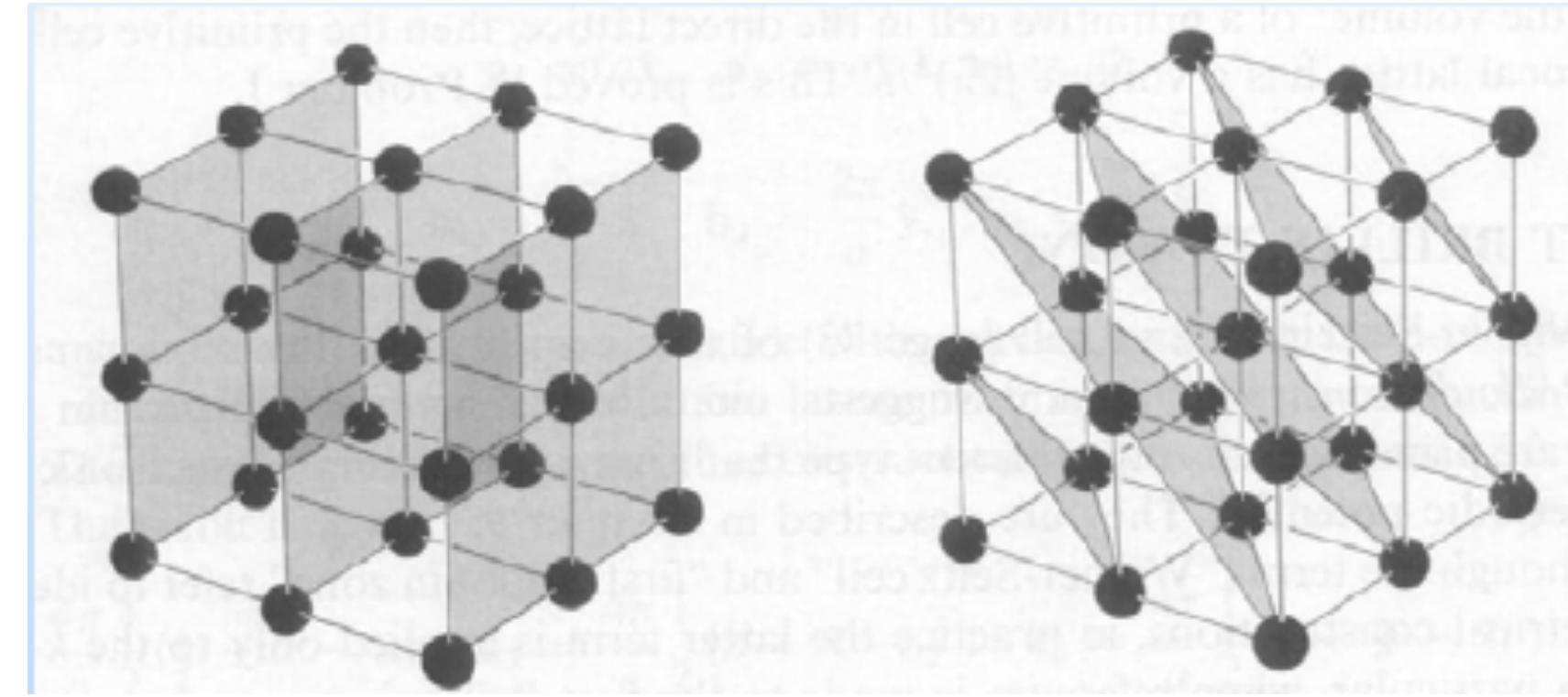


Figure 11.15: Unit cells for All of the Three Dimensional Bravais Lattice Types.

- V = volume of unit cell
- V^* = volume of reciprocal unit cell
- $V V^* = (2\pi)^d$ (d = dimension, 1, 2, 3)
- Reciprocal of fcc is bcc and vice versa
- Reciprocal of hexagonal is hexagonal

[First] **Brillouin zone** $\equiv$ Wigner Seitz Cell of the momentum (i.e. wave vector) space.

- Any plane containing at least three non-collinear Bravais Lattice points $\equiv$ Lattice Plane



- For a given BL and a LP, BL = family of LP
- Any family of lattice planes can be labeled by a reciprocal lattice vector K , which is perpendicular to the lattice planes. The minimum non-zero length of K is given by $2\pi/d$, where d is the spacing between lattice planes.

Any lattice plane is a 2d Bravais Lattice, characterized by two vectors, which we can call **a1** and **a2**.

Now, choose any vector connecting a lattice point of a given lattice plane and another point of the next lattice plane. Call this **a3**. **a1, a2, a3** are primitive vectors.

Let **b1, b2, b3** be their reciprocal primitive vectors. By construction, **b3** is perpendicular to our lattice planes. This is because all 3 rlv are perpendicular to the 3 dlv.

Furthermore, **b3 · a3 = 2π**.

Since d is the lattice plane separation, **b3 · a3 = |b3|d**, and thus **|b3| = 2π/d**. QED.

Miller Indices of a Lattice plane A

We know that families of lattice planes are associated to unique ~~the~~ reciprocal vectors $\vec{K}$. Miller indices of the plane are nothing else but the coordinates of the shortest reciprocal lattice vector perpendicular to a family of planes.

The coordinates of the miller planes are given in a basis of a specific set of primitive reciprocal vectors a plane with miller indices h, k, l is $\perp$ to the reciprocal vector $\vec{K} = h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3$

Say a plane $A \perp \vec{h} = \sum l_i \vec{b}_i$, l_i 's no common factors

$$\text{dist } A_N - A_{N+1} = 2\pi/k$$

The miller indices for $A = (l_1, l_2, l_3)$

Geometrical meaning

A intercepts the axis $\{\vec{a}_i\}$

$$a_1 x_1, x_2, x_3$$

The ec. for A :

$$\vec{h} \cdot \vec{r} = cte.$$

$$\vec{h}(x_1 \vec{a}_1) = \vec{h}(x_2 \vec{a}_2) = \vec{h}(x_3 \vec{a}_3) = 2\pi l_1 x_1 = 2\pi l_2 x_2 = 2\pi l_3 x_3$$

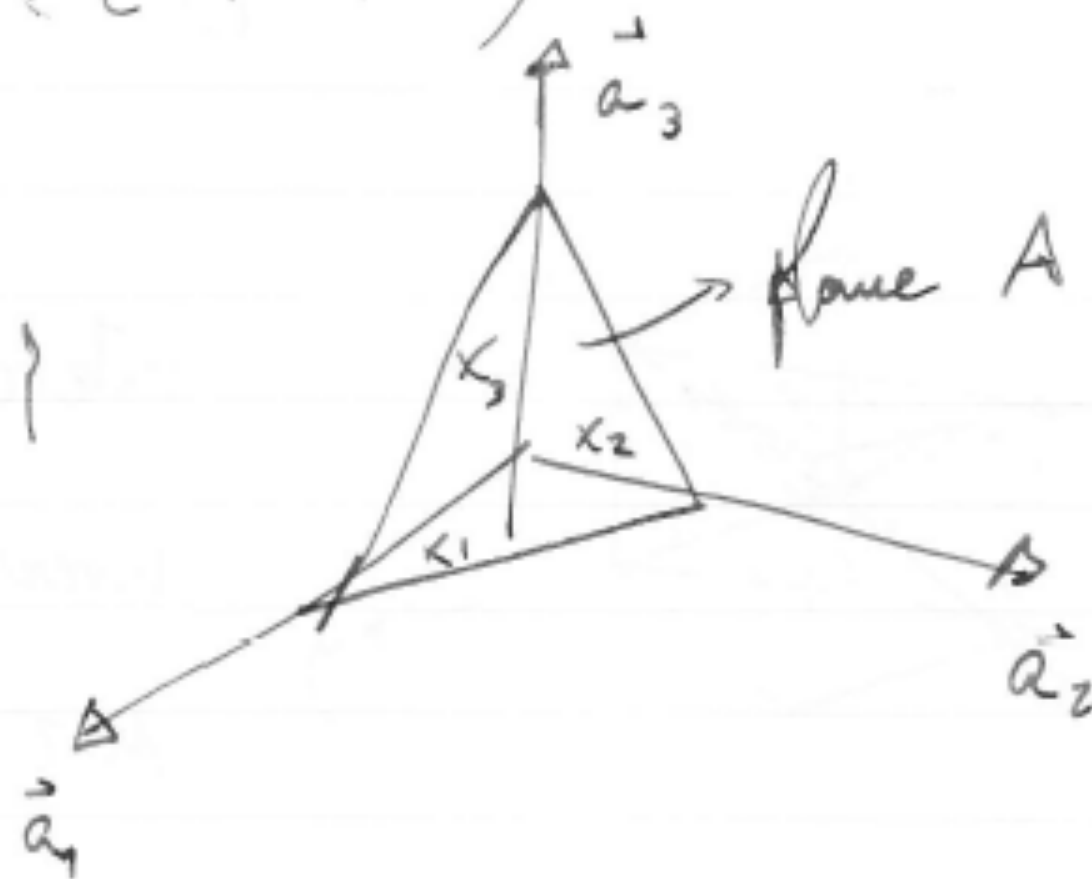
$$l_1 x_1 = l_2 x_2 = l_3 x_3 = cte$$

$$\text{Miller indices} = (l_1, l_2, l_3) \propto \left(\frac{1}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}\right)$$

How to: Determine $x_1, x_2, x_3 \Rightarrow$ where plane intercepts the

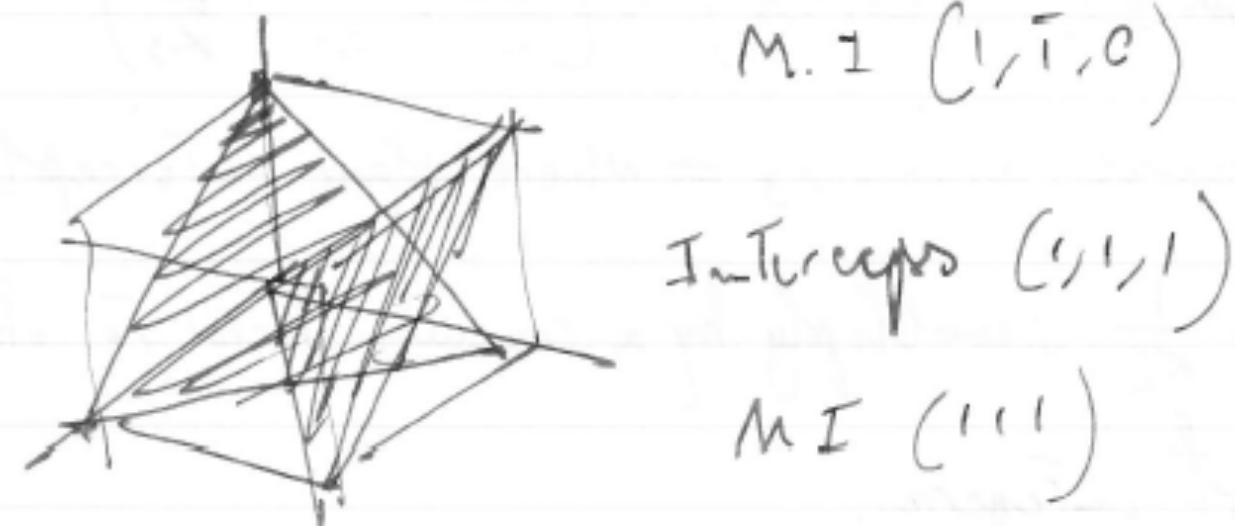
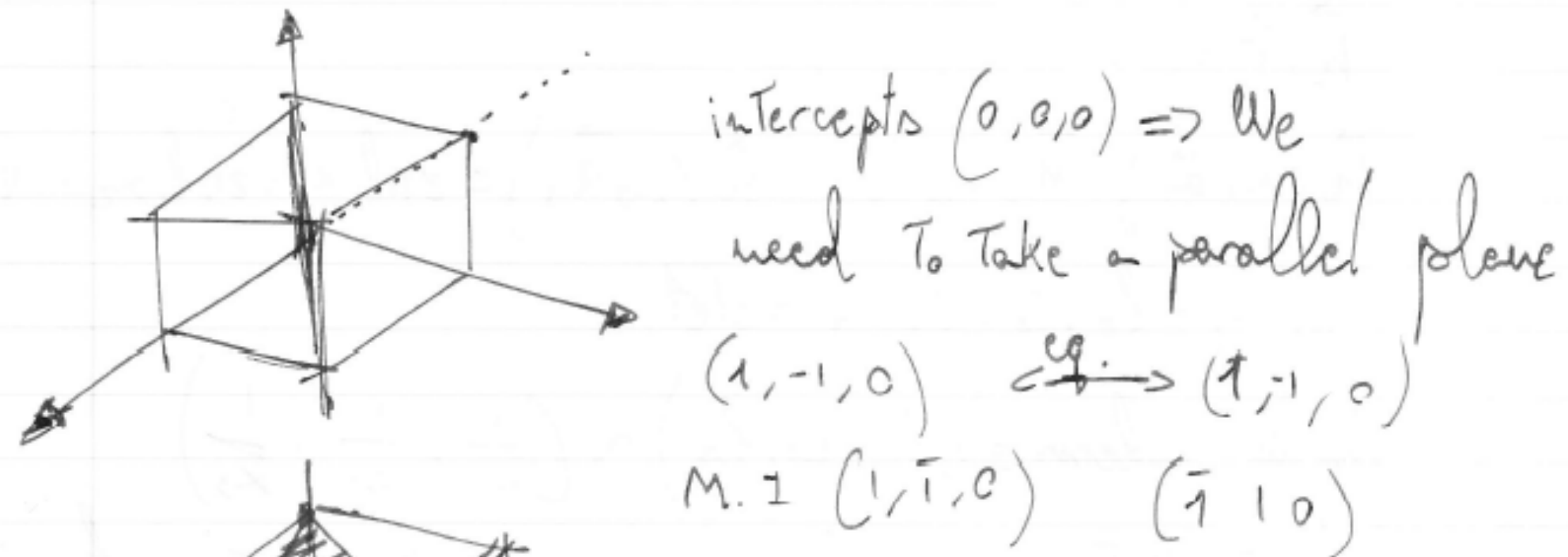
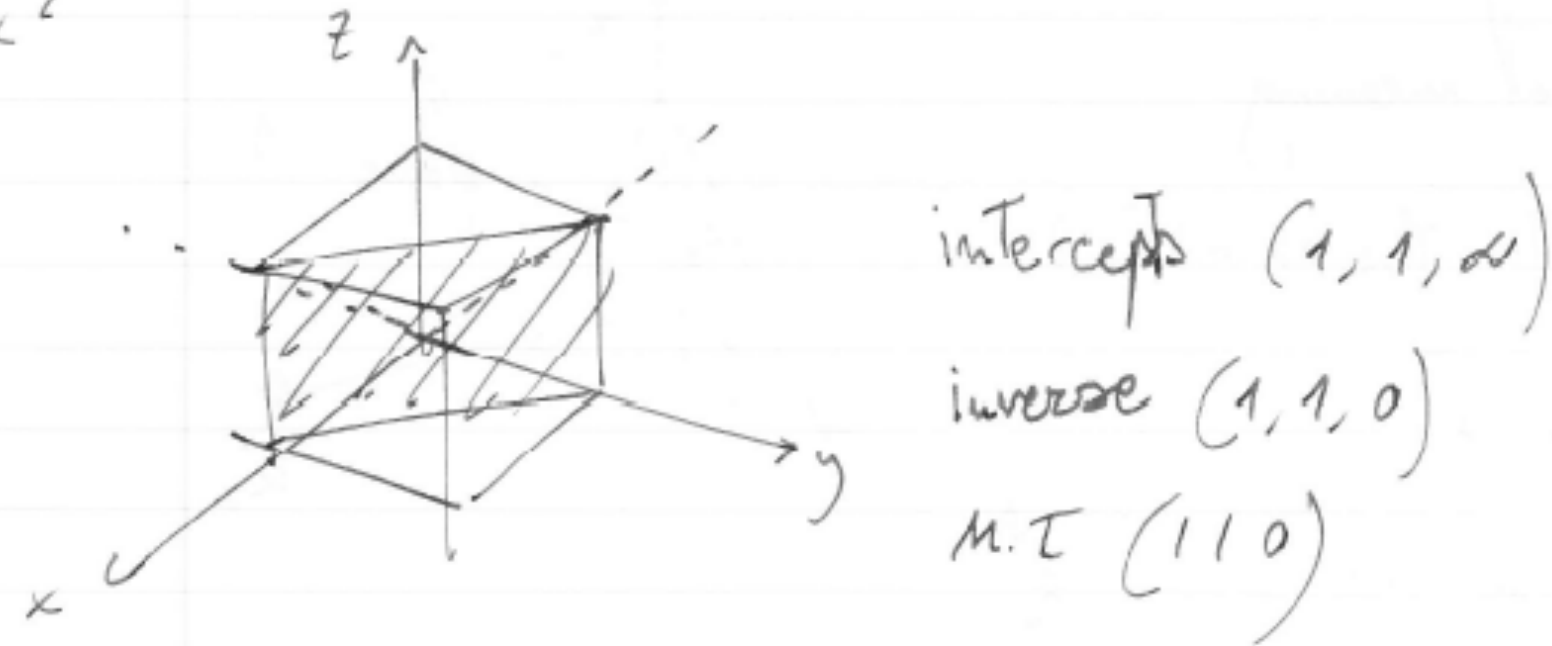
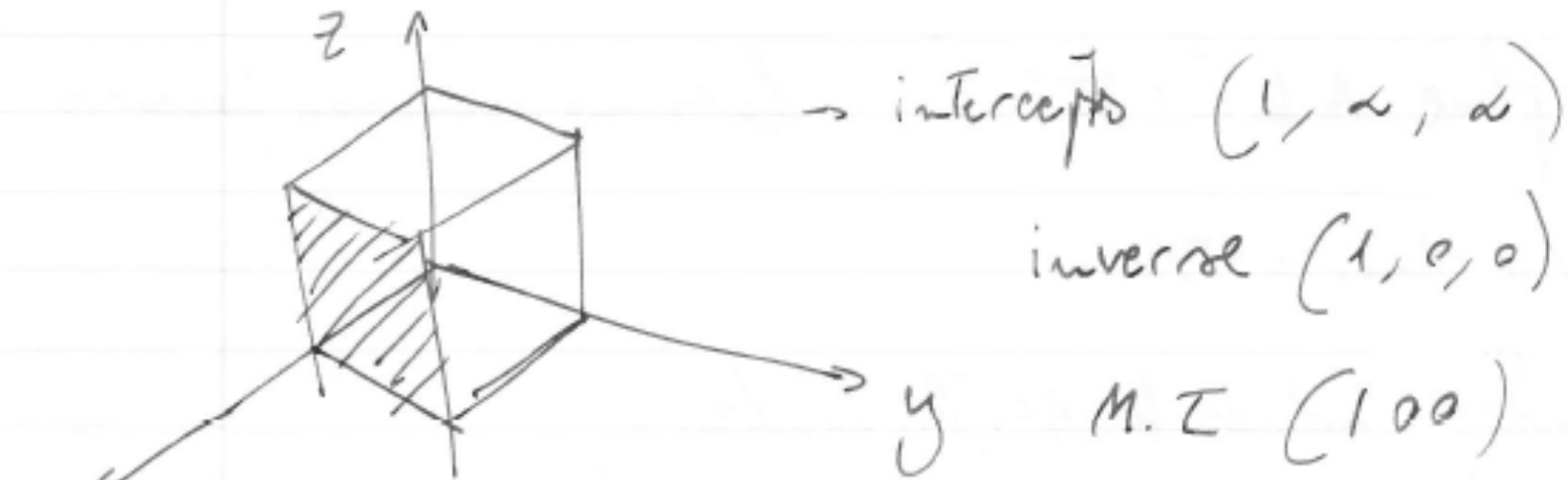
axis; Take $\frac{1}{x_i}$; multiply by a common factor to obtain

the smallest integers.



Conventions to specify directions

1. Usually (for cubic lattices) They are defined with respect to the conventional unit cell



Notations :

(100) a lattice plane

$$(0-10) \equiv (0\bar{1}0)$$

$$\{100\} = (100), (\bar{1}00), (010) \text{ etc}$$

$[100] \equiv$ direction along $\vec{a}_1$

$[xyz]$ direction along $x\vec{a}_1 + y\vec{a}_2 + z\vec{a}_3$

$$\langle 100 \rangle \equiv [100], [\bar{1}00], [010] \text{ etc}$$